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Google and Facebook are gobbling up the internet's subsea cables

Google, Facebook and Microsoft don't just control all our data, now they're taking control of the underwater cables that whizz it around the world

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MORE THAN 5,700 kilometres of the Atlantic Ocean separate the coastlines of France and Virginia. But in 2020 they will be directly connected for the first time by the Dunant submarine cable which will boost internet capacity between Europe and the east coast of the US.

When Dunant becomes operational it will join more than 428 submarine cables, spanning thousands of kilometres, that makeup the backbone of the internet. But this cable is unlike any other.

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Named after the Henry Dunant, the founder of the Red Cross and winner of the first Nobel Peace Prize, the cable is owned by Google. It's the first transatlantic submarine cable that has been privately financed and deployed by one of the big tech firms.

"The issue often for the content providers – Google and Facebook – is they currently cannot source enough capacity from the existing cables for themselves," says Alan Mauldin, research director at TeleGeography, a telecoms data firm that tracks and maps subsea cables. "So they need to build to get that much access themselves."

The result is more of the internet's physical infrastructure is owned by the companies that have the biggest presence online.

Dunant may be the first subsea cables owned by one of the big tech firms to cross the Atlantic Ocean but it isn't the company's first private cable. In January 2016, Google also announced the Curie cable that would run from Chile to Los Angeles. In recent years, Google, Microsoft, and Facebook have all invested in submarine cables to keep pace with increasing demand.

Previously, big technology firms spent millions on cables as parts of consortiums. In return for their investment, each company gets a say in its route and, crucially, a share of its capacity. But big tech doesn't like to share.

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process of building new cables down and that's why Google has decided to go it alone. "Today if you look at the market, the majority of cables are driven mostly by the internet companies."

Facebook's vice president of network engineering, Najam Ahmad says it can take three years to create a new cable, meaning companies and consortiums are planned anywhere up to ten years in advance.

In September 2017, Microsoft, Facebook and Telxius completed the 6,600km transatlantic MAREA cable. It started operating in February 2018 and can transmit data at 160 terabits per second.

Other recent consortium cables involving big tech have hooked up Singapore and Australia and Japan and the US. In total, Facebook has been involved in at least six subsea cable consortiums and Google has interests in at least 13 cables, spanning back to 2010.

Mauldin says that despite Google investing in private cables it will not sell capacity from them for other companies to use. "They would never do that because they would then become a carrier and be subject to licensed as a carrier," he says. ("We don't want to be a service provider as an ISP," adds Google's Hölzle.)

"The cables will not ultimately be used only for Google traffic. What tends to happen is they will swap capacity on this cable with parties with capacity on other cables," Mauldin says. "What effectively happens is you might see that Google has built one cable on a certain route but they can leverage that by using that as a means of exchange."

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All undersea cables are subtly different – in the capacity they can carry and the technologies that are used to build them – but largely work in the same way. They're several inches thick and are comprised of a plastic tube that protects a copper casing with fibre optics inside.

Data is pushed through the cables using light and repeaters spread along the seafloor at distances of around 50 miles, ensuring the data moves along at a consistent speeds. Telegeography's analysis estimates there is more than one million kilometres of subsea cables in use today. One of the longest is the Asia America Gateway cable that stretches around 20,000km.

Initially, submarine cables – the first were laid across the Atlantic in the 1850s – were used for telegraphs and then data needed for mobile phone calls. The cables can last for around up to 25 years and if they're broken (usually by the anchors of ships or undersea earthquakes) they can be repaired by submersible robots.

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Ahmad says Facebook's interest in subsea cables comes from a lack of capacity available. There are two types of Facebook traffic, he adds: machine-to-machine and machine-to-user. The former of these, which involves data

"The capacity that's in the water today is not really sufficient for our projects around the amount of capacity that we need," he says. "The investments in the industry hasn't been keeping up with traffic growth and that's one of the things that we have been helping to drive." Recent subsea cables, such as Havfrue, have been designed to connect countries where more internet capacity is needed. Instead of the Havfrue connecting the US and London, it joins the US to Denmark where cables haven't previously landed.

"The corners of the world that didn't use to have high bandwidth needs actually now do," Hölzle says. "That's why a fair amount of our investment has been in Asia, which has really been going very fast in the last five or ten years."

It shouldn't be a surprise that big technology companies are investing more in submarine cables, Hölzle adds. He says the move is similar to when submarine cables started to be owned by telecoms companies. "The phone companies built their own infrastructure because they needed it and they were the majority users of that infrastructure," he says. "It's just changed that the field has moved from voice to data."

But in the race to control the internet, subsea cables aren't the only solution. Space companies and pioneers, including Elon Musk's SpaceX, are looking to use swarms of small satellites to beam internet from above.

Facebook has trialled using a giant solar-powered drone flying above regions to connect people on the ground. The drone crashed on its second test flight and Facebook has since cancelled the project.

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Separately, Google has spun out Project Loon – its internet-connected balloons that provide connectivity on the ground below – as a separate company. Hölzle says that going forward it would technically be possible (if not practical) for Loon to look at the balloons being used across large networks.

"Subsea cables are kind of what you don't want to build if you can avoid it," Hölzle says."Maybe one-day cables are going to be replaced by that. Light actually travels faster in the air than in fibre. Theoretically, if you had a string of Loons with lasers in-between across the Atlantic it would beat the latency of cables by around one third."

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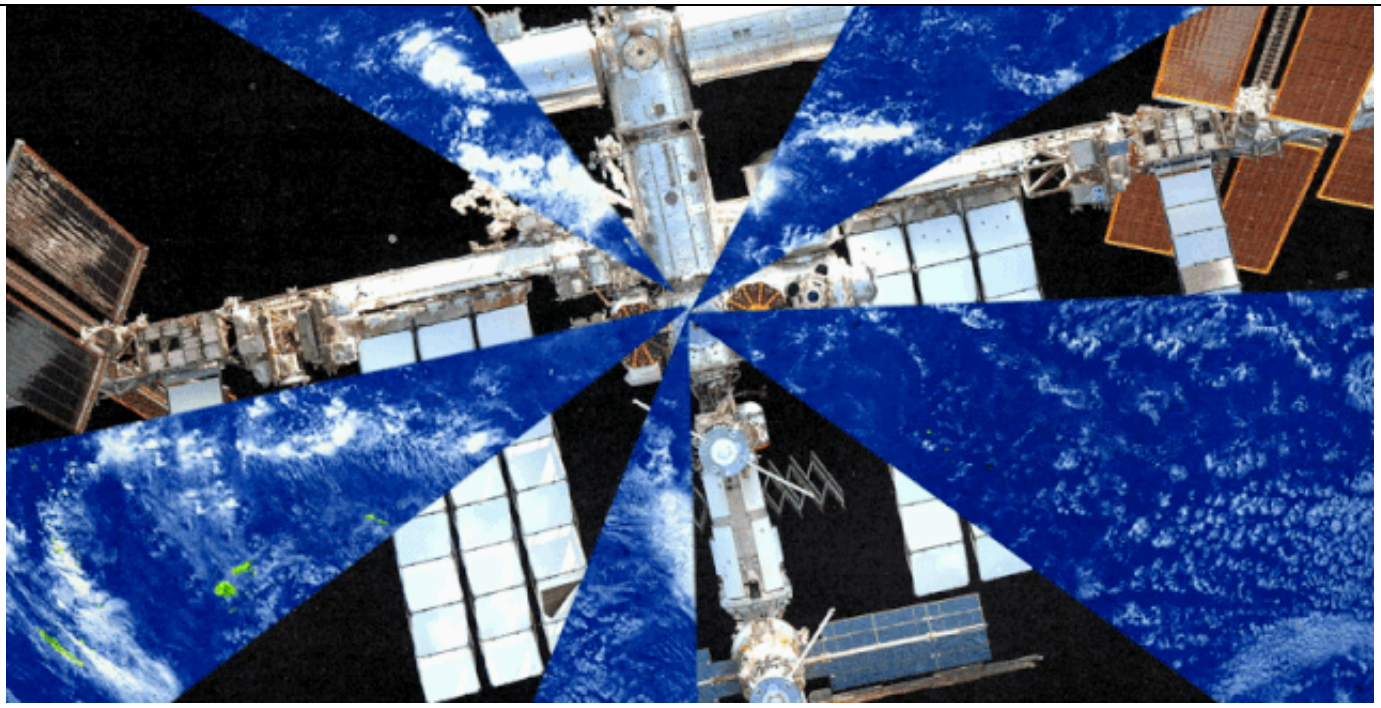

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